Exercise 2: Link State and Distance Vector Routing

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Learning objectives: Apply Dijkstra and Bellman-Ford; construct forwarding tables; analyze convergence, failures, and count-to-infinity.

Network topology:

```
              2
        A ------------ B
        |              |
      5 |              | 3
        |              |
        C ------------ D
         \            /
        1 \          / 2
           \        /
              E
```

Part 1 — Link State Routing

Question 1 — Dijkstra's Algorithm

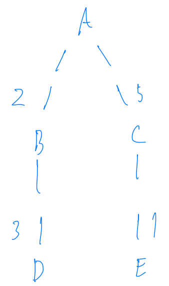
Starting at router A, execute Dijkstra's algorithm. Show every update and the node confirmed at each step.

Step	Confirmed Node	D(B), Prev.	D(C), Prev.	D(D), Prev.	D(E), Prev.
0	A	2 ,A	5, A	∞ , -	∞, -
1	B	2, A	5, A	5 ,B	∞ , -
2	C	2, A	5, A	5, B	6 ,C
3	D	2,A	5, A	5,B	6, C
4	E	2, A	5, A	5, B	6, C

Question 2 — Shortest-Path Tree

Draw the shortest-path tree rooted at A, then list the shortest path and total cost.

SPT sketch:



Destination	Shortest Path	Total Cost
B	A→B	2
C	A→C	5
D	A→B→D	5
E	A→C→E	6

Question 3 — Forwarding Table

Destination	Shortest Cost	Next Hop	Complete Path
B	2	B	A→B
C	5	C	A→C
D	5	B	A→B→D
E	6	C	A→C→E

Part 2 — Distance Vector Routing

Consider the same network. Initially, each router knows only the cost to its directly connected neighbors.

```
             2
      A ------------ B
      |              |
    5 |              | 3
      |              |
      C ------------ D
       \            /
      1 \          / 2
         \        /
            E
```

Question 4 — Bellman-Ford Update

Router A receives the following distance vectors from its neighbors:

From B:

A=2, B=0, C=4, D=3, E=5;

From C:

A=5, B=4, C=0, D=1, E=1.

Use the Bellman-Ford equation. Calculate router A's new distance to:

Show all calculations, including the neighbor through which the minimum-cost route is obtained.

Destination	Via B calculation	Via C calculation	Minimum Cost	Next Hop
B	2+0 = 2	5+4=9	2	B
C	2+4 =6	5+0 =5	5	C
D	2+3 =5	5+1 =6	5	B
E	2+5 =7	5+1 =6	6	C

Question 5 — Distance Vector Table

After receiving the distance vectors, construct router A's updated routing table.

Destination	Next Hop	Distance	Full Path
B	B	2	A+B
C	C	5	A→C
D	B	5	A→B→D
E	C	6	A→C→E

Part 3 — Routing Convergence

Question 6 — Link Failure

The C–D link fails. Explain separately what happens with Link State and Distance Vector routing.

Link State routing.
Your answer should describe:

- Which router(s) detect the failure.
- What information is flooded through the network.
- How routers update their link-state databases.
- How routers calculate new shortest paths.

Distance Vector routing.
Your answer should describe:

- How routers learn about the failure.
- How updated distance vectors propagate.
- Why routers may temporarily have incorrect routing information.
- How routing eventually converges.

Link State	Distance Vector
Detect Failure	Detect failure
Flood LSA	ส่ง distance vector ให้ neigbors
Update LSDB	Update distance table
Run Dijkstra	Run Bellman-ford
Converges relatively fast	อาจ converge ช้ากว่า

Part 4 — Count-to-Infinity

Consider

```
      1          1
A ------- B ------- C
```

Initially A reaches C through B at cost 2. The B–C link fails, but B receives stale information from A saying C is reachable at cost 2.

Question 7

1. Why can this create a routing loop? เพราะ B คิดว่า A ยังไป C ได้ ทั้งที่ เส้นเชื่อม B-C ไปแล้ว

2. Show how the estimated distance to C can increase over successive iterations. A - B = 1 เพราะ A → C = 2 มันขึ้นแบบ 2→3→4→5→...

3. What is count-to-infinity? router ยังได้รับข้อมูลผิด หรือเก่าจาก neighbor

4. Why does it occur in Distance Vector routing? เมื่อ link เสีย router จะยังเชื่อข้อมูลเก่าจาก neighbor เลย เกิด routing loop ได้

5. Name a mitigation approach.

Split Horizon, Route Poisoning, Poison Reverse, Hold-down Timer

Part 5 — Compare LS and DV

Question 8

Characteristic	Link State	Distance Vector
Basic algorithm	Dijkstra	Bellman-Ford
Information maintained	Link-state database / topology	Distance vector / routing table
Information exchanged	Link-state information / LSA	Distance vectors
Algorithm	Shortest Path First	Bellman-ford
Topology knowledge	รู้ topology ของ network	รู้ระยะทางผ่าน neighbors
Routing updates	flood ไปทั่ว network	ส่งให้ neighbors
Convergence	โดยทั่วไปเร็ว	โดยทั่วไปช้ากว่า
Routing loops	มีโอกาสน้อยกว่า	มีโอกาสเกิดได้
Count-to-infinity	ไม่มีปัญหาเหมือน DV	เป็นปัญหาสำคัญ
Example protocol	OSPF, IS-IS	RIP

Part 6 — Design Question

Question 9

You are designing a routing system for an enterprise with 500 routers requiring fast convergence, accurate topology knowledge, efficient routing, and hierarchical design. Choose LS or DV and justify your choice with at least three technical arguments. Your answer should discuss at least three advantages and/or disadvantages of your chosen approach.

I would choose Link state routing

1. Fast convergence

2. Accurate topology knowledge

3. Efficient routing

4. Scalability และ Hierarchical design

5. ลดปัญหา routing loop และ count-to-infinity

Challenge — Algorithmic Analysis

Question 10

For N routers and E links, compare LS and DV in information maintained, communication overhead, computation, convergence, and scalability. Explain why large-scale intra-domain routing generally favors Link State.

Large-scale network เหมาะกับ Link state เพราะ Convergence เร็วกว่า

, topology แม่นยำ และรองรับ hierarchical design ได้ดี